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20th July 2026.

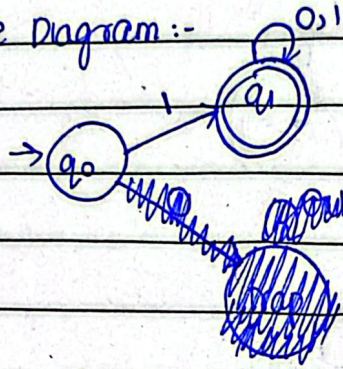
Date: _____

ITC Assignment 4 - Harris Shibli (28988)

Qs1. A. Regular Expression:-

$1(0+1)^*$

B. DFA State Diagram:-



C. Test Strings:-

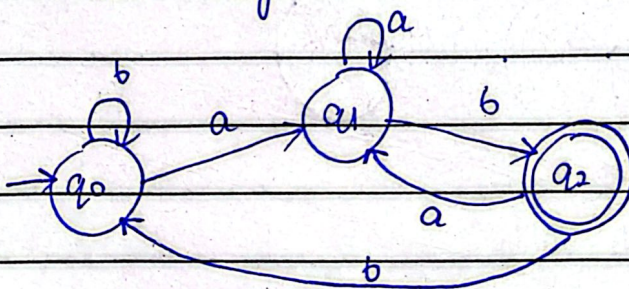
Accepted : 1, 101 ie must end in an accept state.

Rejected : ~~0, 011~~ 0, 011 (any string starting with 0's)

Qs2. Ends with "ab" $\Sigma = \{a, b\}$

A. $(a+b)^*ab$

B. DFA State Diagram:-



C. Test Strings:

Accepted : ab, bab, aab

Rejected : a, aba, abaa

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Harris Shibli
(28988)

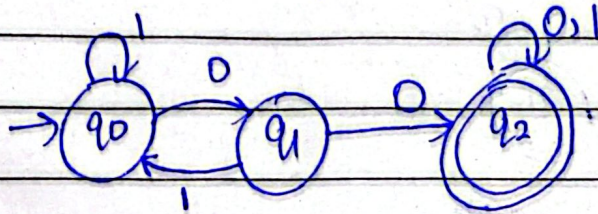
Date: _____

Qs3. Contains "00" $\Sigma = \{0,1\}$

A. Regular Expression:

 $(0+1)^* 00 (0+1)^*$

B. DFA State Diagram:-



C: Test Strings:

Accepted: 00, 1001, 1000

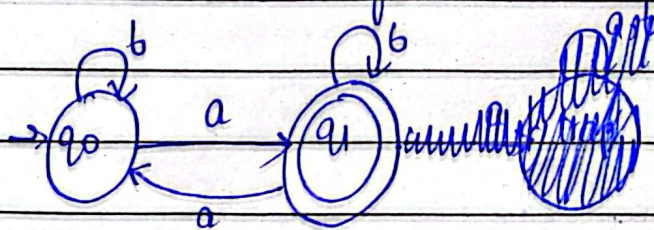
Rejected: 0, 0101, 101

Qs4. Exactly one 'a' ($\Sigma = \{a,b\}$)

A. Regular Expression:

 $b^* ab^*$

B. DFA State Diagram:



C: Test Strings:

Accepted: a, bab, babb

Rejected: bb, aab, bbb, baa

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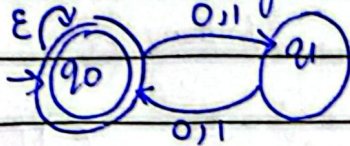
Date: _____

Q5. Even length $\Sigma = \{0,1\}$

A. Regular Expression:

$((0+1)(0+1))^*$

B. DFA State Diagram:-



C. Test Strings

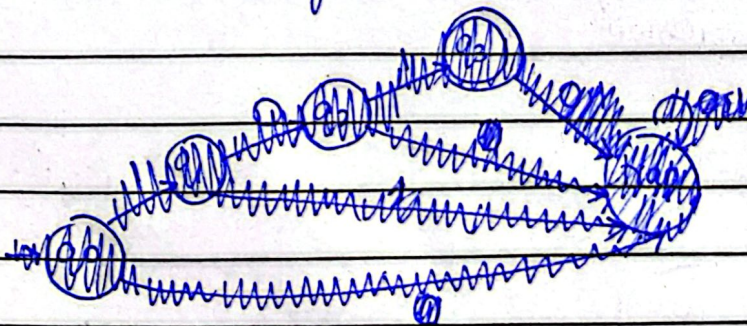
Accepted: $\epsilon, 00, 01$

Rejected: $0, 011$

Q6. Exactly "101" $\Sigma = \{0,1\}$

A. Regular Expression: 101

B. DFA State Diagram:-



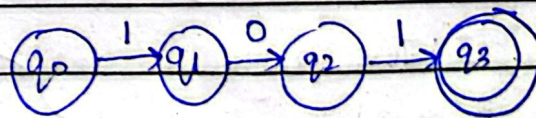
Redrawn below:

C. Test Strings:

Accepted: 101

Reject: 1, 10, 1010

↳ any string apart from 101



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Haria Shibli (28988)

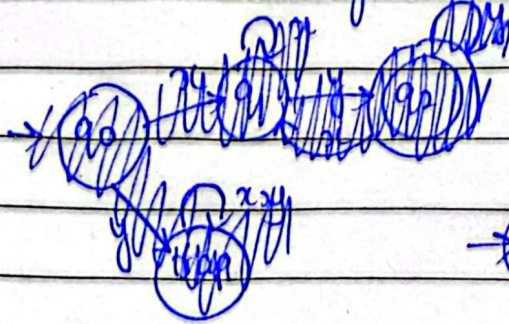
Date: _____

Qs 7. Starts with x AND ends with y ($\Sigma = \{x, y\}$)

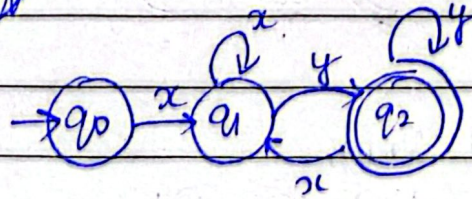
A. Regular Expression

$x(x+y)^*y$

B. DFA State Diagram



Redrawn:



C: Test Strings

Accepted: xy, xxy

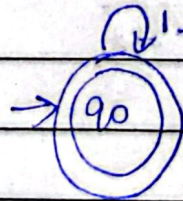
Rejected: x, yxy

Qs 8. All 1's including ϵ ($\Sigma = \{0, 1\}$)

A. Regular Expression

1^*

B. DFA State Diagram



C: Test Strings

Accepted: $\epsilon, 1, 11, 111$

Rejected: $0, 110$

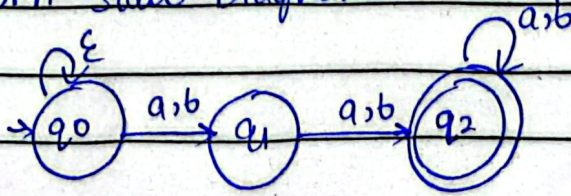
→ It can contain any combination of 1's including ϵ .

Qs 9. length ≥ 2 $\Sigma = \{a, b\}$

A. Regular Expression

$(a+b)(a+b)(a+b)^*$

B. DFA State Diagram



C. Test Strings:

Accepted: aa, aba

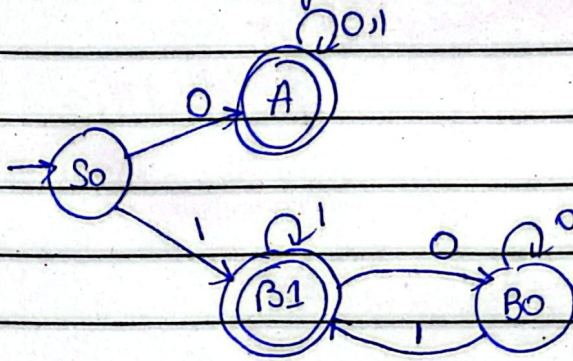
Rejected: ϵ, a

Qs 10. Starts with 0 OR ends with 1 ($\Sigma = \{0, 1\}$)

A. Regular Expression

$0(0+1)^* + (0+1)^*1$

B. DFA State Diagram:



C. Test Strings

Accepted: $01, 11$

Rejected: $10, 100$